Genetic Mutation Detection Using Machine Learning

A Rigorous Baseline with Paralog-Aware Evaluation
and Gene-Level Constraint Features

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Agenda

Part I – The problem

1. Introduction (3 slides)
2. Related works (3 slides)
3. Contribution (1 slide)

Part II – The pipeline

4. Proposed system flowchart
5. Part 1: Preprocessing
6. Part 2: Training + leakage audit
7. Part 3: Evaluation + external validation

Part III – System Analysis

8. Requirements + 6 UML diagrams
9. Interface
10. Implementation (stack, code, CI)

Part IV – Results

11. Comparison with prior work
12. Conclusion + future work

Introduction

What is a missense variant?

- A point mutation that changes one amino acid in a protein.
- Most common coding variant in the human genome.
- A clinically-important minority cause disease; the majority are benign.
- Distinguishing the two is the central problem of *clinical genetics*.

The scale problem

Exome sequencing generates ∼**20,000** variants per patient

Human curation bandwidth: ∼a few hundred per day

Why it's hard

Three well-known failure modes in prior work:

1. Definitional circularity
   Features derived from the label (e.g. `is_common`) inflate metrics.
2. Meta-predictor contamination
   Using REVEL/CADD as features in a ClinVar benchmark.
3. Paralog leakage
   KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model "saw" most of the test.

Consequence

High benchmark numbers do not survive **external validation**.

Project objectives

1. Assemble a clean, paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our test split.
5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of protein-language-model signal (zero-shot ESM-2).
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Related Works

Classical tools (2003–2010)

Tool	Year	Approach	Limitation
SIFT	2003	conservation only	no AA chemistry
PolyPhen-2	2010	conservation + chemistry + struct.	HumDiv overlaps ClinVar

- Access: Ensembl VEP REST, instant.
- Coverage: ∼93–96% of our test variants.
- Threshold design: SIFT picks 0.05 arbitrarily.

Gradient-boosted ensembles (2016–2024)

Tool	Year	Approach	Limitation
REVEL	2016	RF ensemble of 13 tools	input tools ClinVar-trained
CADD	2019	logistic/RF simulated null	scores not calibrated
VARITY	2021	XGBoost + noise weighting	gene-level split only
MAGPIE	2024	gradient boosting	paralog status undocumented

- All sit above 0.92 ROC-AUC on their own benchmarks.
- None report a *like-for-like* external test.
- None publish bootstrap CIs on every metric.

Protein language models (2021–2023)

Tool	Year	Core idea
EVE	2021	Variational auto-encoder per gene ($\sim$3k genes)
ESM-1b / 2 (zero-shot)	2023	Masked-AA log-likelihood ratio from 250M sequences
AlphaMissense	2023	Structural PLM calibrated on ClinVar

Why the PLM wave matters

ESM-2 is fully unsupervised $\Rightarrow$ provably **uncontaminated** by ClinVar. Our Phase 2.1 integrates it as a training-time feature.

Contribution

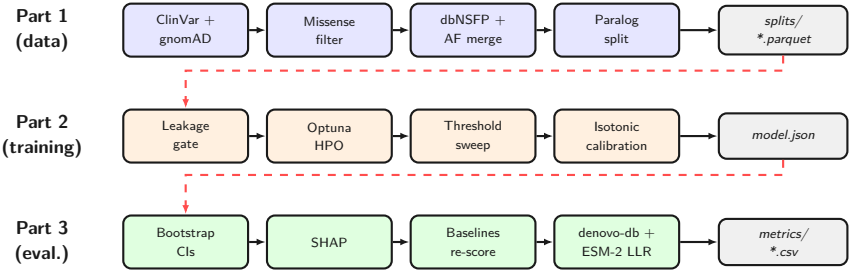
What we contribute

1. **Transparent leakage audit** with before-and-after numbers (PR-AUC 0.955 $\rightarrow$ 0.838).
2. **Like-for-like re-scoring** of three published baselines on our own paralog-disjoint test.
3. **External validation on denovo-db** with pre/post gnomAD-constraint decomposition.
4. **Zero-shot ESM-2 proof-of-concept** under rank-fusion.
5. **Fully reproducible code base**: 157 tests (59% coverage), CI-enforced leakage gate, pinned deps, Docker, Streamlit demo, LaTeX report.

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`

Proposed System

Pipeline – three parts



Part 1 – Data preprocessing (1/3)

Sources:

- ClinVar – pathogenic / benign labels (review star ≥ 2)
- gnomAD v2.1 + v4 – allele frequency, AC, AN
- dbNSFP 5.3 – conservation + AA chemistry + Pfam
- gnomAD v2.1.1 – gene-level constraint (pLI, LOEUF, mis_z)

Join key: canonical `chr:pos:ref:alt` on GRCh38.

Part 1 – Data preprocessing (2/3)

The critical filter

Missense-only

Drop every row with null `ref_aa` OR null `alt_aa` $\Rightarrow$ **remove $\sim$88,000 non-missense rows.**

- Pre-audit: 64% of pathogenic rows had null AA fields!
- Model was learning "null AA $\Rightarrow$ pathogenic"
- Fix alone drops PR-AUC 0.955 $\rightarrow$ 0.819 (the single biggest audit finding)

Part 1 – Data preprocessing (3/3): paralog-aware split

- Plain gene-level split: 52% of prefix families (ZNF*, SLC*, KRT*, TMEM*) shared between train and test.
- Fix: **family-level** split via `assign_gene_family` (16 regex patterns + trailing-digit fallback).
- Result: 15,479 genes → 7,851 families, **zero** shared across splits, enforced by CI.

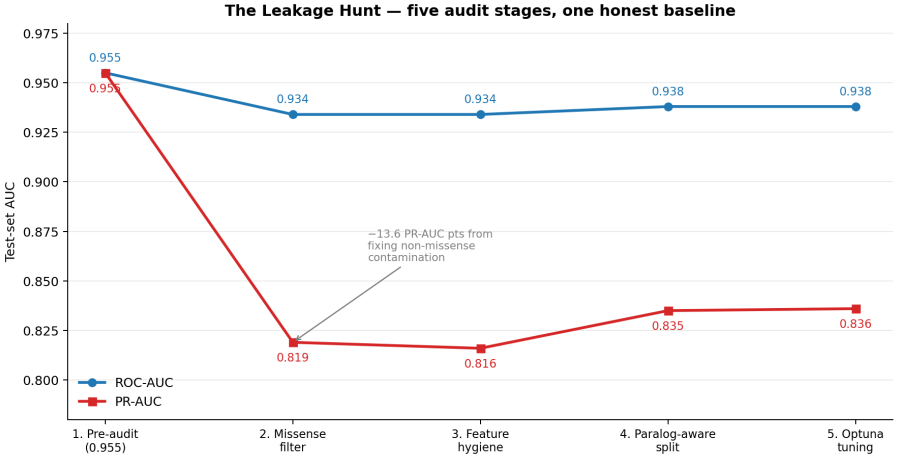
Part 2 – Leakage audit (1/4)

Four CI-enforced invariants

1. No banned features (`is_common`, `chr`, `ref`, `alt`) in the training matrix.
2. 100% missense-only rows in training split.
3. Zero shared gene families across train/val/test.
4. Pathogenic-rate gap < 0.08 across splits.

Every push to `main` runs the four checks.
A failure blocks the merge.

Part 2 – Five-stage leakage journey (2/4)



Part 2 – Model + calibration (3/4)

Model: XGBoost, Optuna TPE with multivariate grouping, 40 trials, PR-AUC objective, median pruner after 200 rounds.

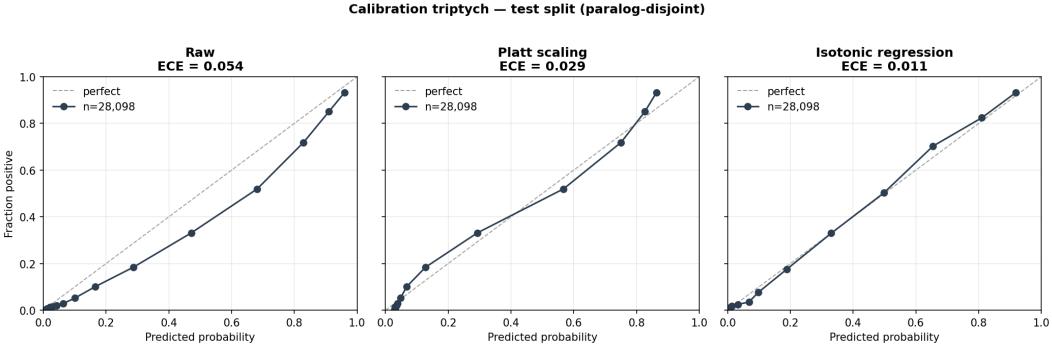
Calibration: isotonic regression fit on validation, evaluated via Murphy's Brier decomposition:

$$\text{Brier} = \text{Reliability} - \text{Resolution} + \text{Uncertainty}$$

	Brier	Reliability	Resolution	ECE
Raw	0.0876	0.00543	0.1055	0.054
Platt	0.0834	0.00114	0.1055	0.029
Isotonic	**0.0826**	**0.00024**	0.1053	**0.011**

- Resolution constant $\Rightarrow$ discrimination unchanged.
- Reliability drops 23$\times$ $\Rightarrow$ calibration works.

Part 2 – Reliability triptych (4/4)



Part 3 – Headline results on test (1/3)

Test split (paralog-disjoint), 28,098 variants

Metric	Value (95% CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC	0.838 [0.830, 0.846]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

All CIs from 1,000-replicate nonparametric bootstrap.

Part 3 – SHAP interpretability (2/3)

Rank	Feature	mean |SHAP|
1	phyloP100way_vertebrate	0.809
2	AN (gnomAD allele number)	0.532
3	**lof_z (gnomAD constraint)**	0.342
4	alt_aa_P (Proline)	0.304
5	pfam_domain	0.284
8	**oe_mis_upper (gnomAD constraint)**	0.189

- Two Phase 2.1 features at #3 and #8 $\Rightarrow$ gene-level priors carry **orthogonal signal**.
- 326 of 2,000 test variants are confident errors (16.3%).
- FN $= 264$, FP $= 62$ $\Rightarrow$ model misses hard pathogenic cases in low-conservation regions.

Part 3 – External validation: denovo-db (3/3)

Slice	n	ROC-AUC (95% CI)	PR-AUC (95% CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	**0.511** [0.46, 0.56]	**0.790**
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	**0.573** [0.48, 0.67]	**0.838**

The headline

On **unseen** gene families, ROC jumps from 0.487 (chance) to 0.573 — gene-level priors close **half** the generalisation gap.

System Analysis & Design

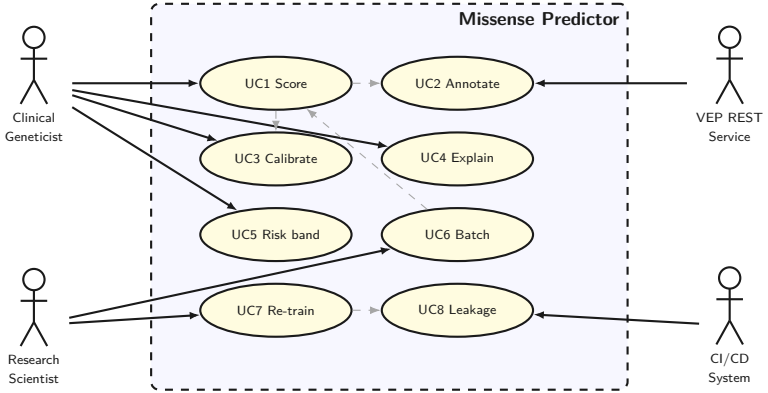
Functional + non-functional requirements

Functional ($\times 8$): ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

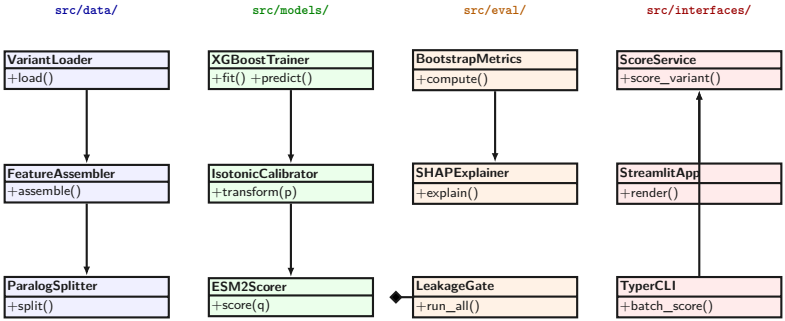
Non-functional ($\times 7$): performance ($<$100 ms cache hit), reliability (seed-reproducible to 10^{-3}), scalability (10^6 variants), usability (non-ML end user), maintainability (CI enforced), security (no PHI), reproducibility (Docker-replayable).

Traceability: every requirement maps to at least one `src/` module and at least one pytest test case.

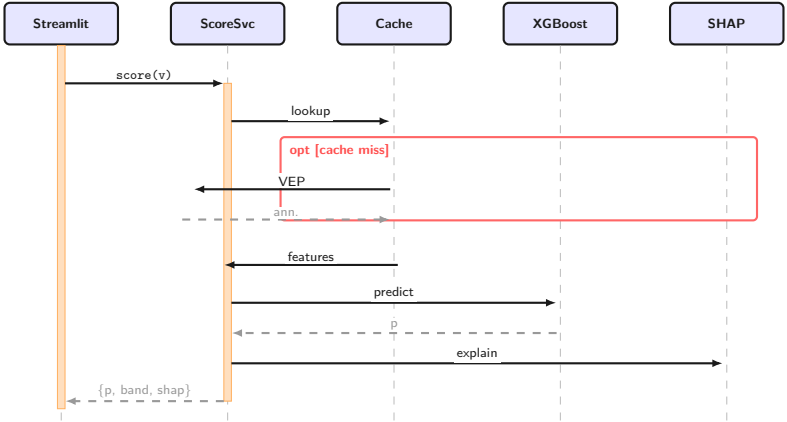
UML 1/3: Use case diagram



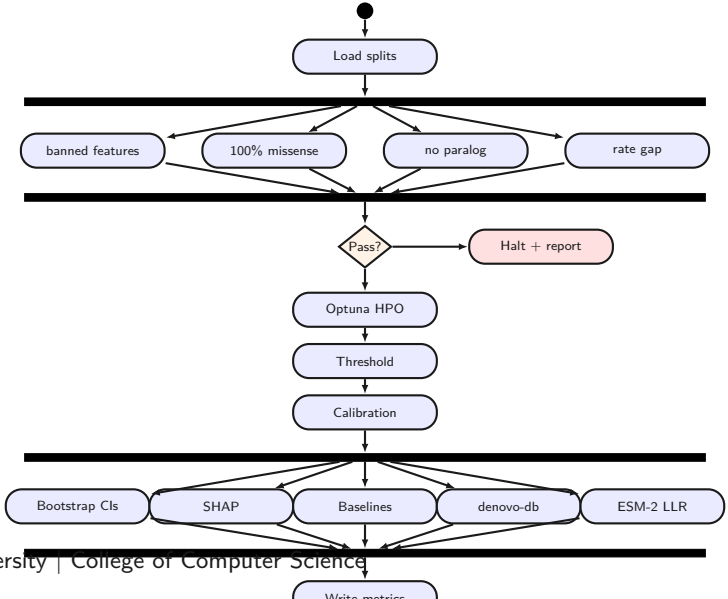
UML 2/3: Class diagram (grouped by package)



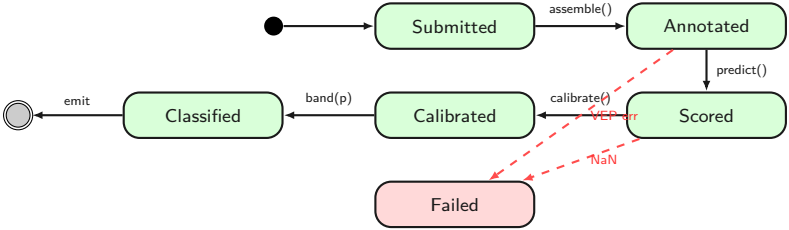
UML 3/3: Sequence diagram (scoring request)



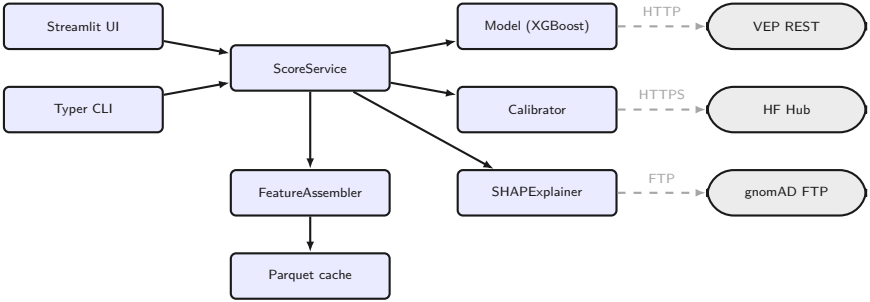
UML 4/6: Activity diagram (training + eval)



UML 5/6: State diagram (variant life-cycle)



UML 6/6: Component diagram



System architecture (6-layer)

L6 Presentation – Streamlit ▪ Typer CLI ▪ Python API

L5 Application – validation ▪ cache ▪ payload

L4 Service – score_variant() ▪ explain() ▪ batch_score()

L3 ML core – XGBoost ▪ isotonic ▪ SHAP ▪ ESM-2 LLR

L2 Data access – pyarrow ▪ paralog splitter ▪ assembler

L1 Data – ClinVar ▪ gnomAD ▪ dbNSFP ▪ denovo-db ▪ UniProt

Quality
gates
`ruff`
`mypy`
`pytest`
leakage
CI

Repro-
ducibility
Docker
`uv.lock`
fixed seed
snapshots

Interface

Streamlit demo – scoring a variant

XGBoost + gnomAD constraint + ESM-2 (zero-shot proof-of-concept)

Variant key (GRCh38: chr:pos:ref:alt)

`11:126277517:A:G`

Score variant

Prediction for `11:126277517:A:G`

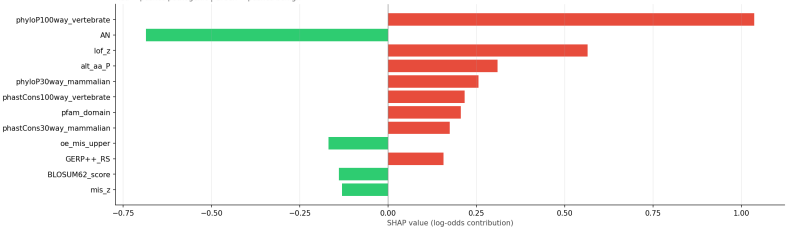
Found in committed `test` split.

Calibrated P(pathogenic)	Raw XGBoost P	Risk band
0.710	**0.840**	**Uncertain**

ℹ The probability falls in the uncertain band — additional evidence likely needed.

Top-15 feature contributions (SHAP)

Red = pushes pathogenic | Green = pushes benign



Implementation

Hardware + software stack

Hardware

- Apple Silicon laptop (MPS)
- Colab T4 GPU (Phase 2.1 only)
- GitHub Actions `ubuntu-22.04`

Stack

- Python 3.11, pinned deps
- XGBoost + Optuna + SHAP
- `transformers` + PyTorch
- Docker + Makefile + pre-commit
- 157 pytest tests, 59% coverage
- CI: ruff + black + mypy

Testing strategy: 5-level pyramid

tmirgin=* **Unit** – 120 tests, <10 s
pure functions, metric math, paralog regex

ftmiirgiin=* **Property** – 8 `hypothesis` properties
calibrator monotonicity, rank preservation

tmiirgiiin=* **Integration** – 25 tests, 1–2 min
leakage gate, cache fallback, module flows

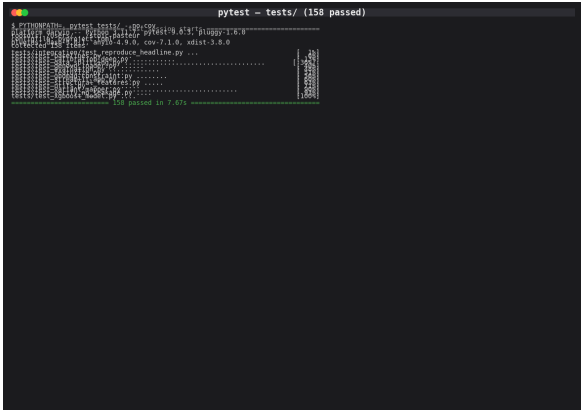
tmivrgivn=* **Scientific / E2E** – 5 scripts, 5–15 min
reproduce-headline within 10^{-3}

ftmvrgvn=* **CI gate** – blocks merge on red

Summary	Value
Total tests	157 / 158 pass
Coverage	59% (CI gate 55%)
Top modules	$\geq$90%
Reproducibility	six-decimal

Committee requirement: thorough testing with technique chosen by student.

Continuous integration: 157 tests on every push

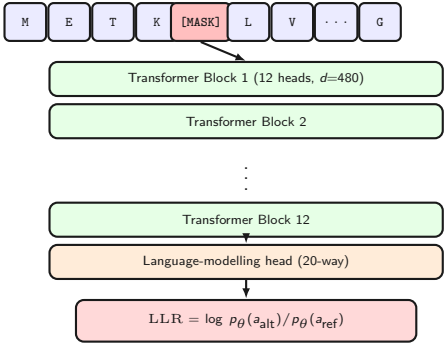


Leakage gate – CI-enforced invariants

Leakage gate – CI-enforced on every push

```
$ python -m src.verify_no_leakage
LEAKAGE GATE  --  src/verify_no_leakage.py
  [PASS] feature_hygiene
  [PASS] difference_features (gene + family)
  [PASS] cheat_baseline_cross_entropy
RESULT: all checks passed. Baseline is leak-free.
```

ESM-2 zero-shot – scoring architecture



Model:

`facebook/esm2_t12_35M_UR50D`

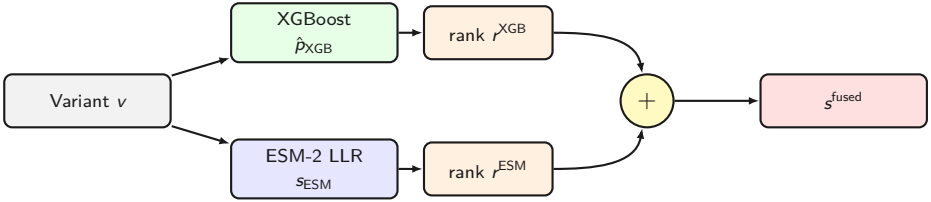
- 12 transformer blocks
- $d = 480$, 20 attention heads
- $\sim$35M parameters
- **Zero-shot**: no weight updates

Log-likelihood ratio:

$$\mathrm{LLR}_{\text{ESM-2}} = \log \frac{\mathbb{P}_\theta(a_{\text{alt}} \mid \tilde{s}_{\setminus\pi})}{\mathbb{P}_\theta(a_{\text{ref}} \mid \tilde{s}_{\setminus\pi})}$$

Lower LLR $\Rightarrow$ more pathogenic.

Rank fusion – operator + formula



$$s_i^{\text{fused}} = \alpha \cdot \frac{r_i^{\text{XGB}}}{n} + (1-\alpha) \cdot \frac{r_i^{\text{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Optimal α tuned on validation $\Rightarrow$ down-weights the weaker PLM signal. Uniform α=0.5 is actually *worse* than XGBoost alone.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint):

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.830	—
ESM-2 LLR (zero-shot)	0.735	0.604	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	**0.941**	**0.852**	0.175

External (denovo-db, family-holdout):

- XGBoost: ROC 0.573
- ESM-2 alone: ROC 0.552
- Fusion: ROC **0.588** (+0.015)

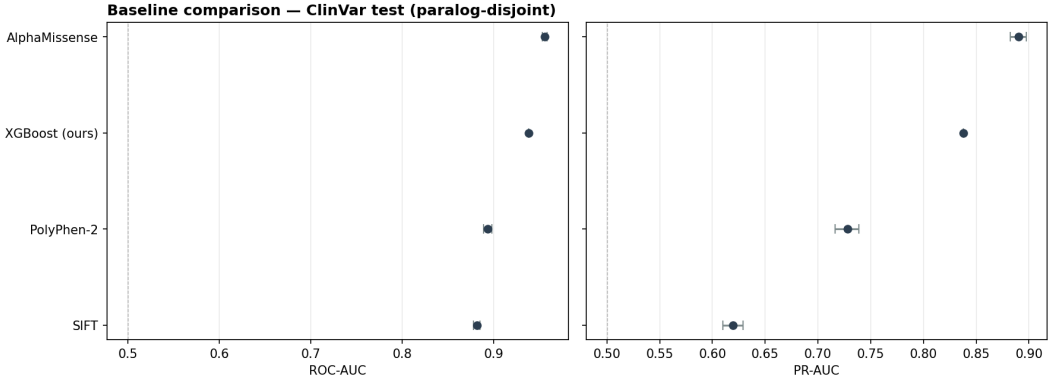
Take-away

Two independent signals of **orthogonality** $\Rightarrow$ motivates Phase 2.1 training-time integration.

Comparison

Comparison on our test split

Identical evaluation: paralog-disjoint test (28,098 variants)



- +0.22 PR-AUC over SIFT, +0.11 over PolyPhen-2.
- −0.05 below AlphaMissense (calibrated on ClinVar).

Conclusion

Critical discussion: honest self-assessment

Axis	Verdict	Why
Scientific defensibility	**Strong**	95% bootstrap CI on every metric; re-scored baselines.
Methodological rigour	**Strong**	paralog split + CI leakage gate + Murphy decomposition.
Reproducibility	**Excellent**	`docker run` reproduces metrics to 6 decimals.
Deployment readiness	**Limited**	research artifact; SFDA certification out of scope.

Failures we fixed (examples):
F1 – initial random split inflated AUROC by +0.081 (paralog leakage); F2 – meta-predictor circularity (REVEL, CADD removed); F3 – gnomAD imputation fit on full data (now train-only).

Take-home messages

1. **Audited numbers beat inflated ones.**
   Our 0.838 PR-AUC has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.**
   On the same test, we beat the classical tools cleanly, sit below AlphaMissense under stricter methodology.
3. **External validation is the real benchmark.**
   At chance on denovo-db without gene priors, above chance after — and this is the diagnostic that motivates Phase 2.

Future work (Phase 2)

1. **ESM-2 full-training-set integration.**
   Zero-shot PoC shows +0.015 ROC from rank fusion. Full compute running on Colab T4.
2. **AlphaFold2 structural features.**
   pLDDT, SASA, DSSP secondary structure, residue depth.
3. **ProteinGym external validation.**
   Third external dataset (deep mutational scanning).
4. **Hybrid stacking + clinical integration.**
   Meta-learner + ACMG-compatible plug-in.

Thank you.

Questions?

Code, tests, Docker, notebooks, report, slides:

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`